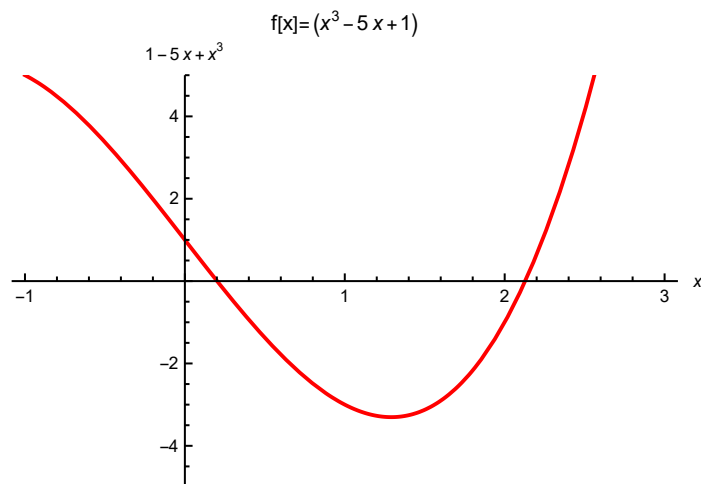


Practical1(Bisection Method)

Question 1

```
In[*]:= f[x_] := x^3 - 5 * x + 1;  
Plot[f[x], {x, -1, 3}, PlotRange -> {-5, 5},  
PlotStyle -> {Red, Thick}, PlotLabel -> "f[x] = " f[x], AxesLabel -> {x, f[x]}]
```

Out[*]=



```
In[*]:= x0 = 0;  
x1 = 1;  
Nmax = 20;  
eps = 0.0001;  
If[N[f[x0] * f[x1]] > 0,  
Print["your value do not satisfy the IVP , so change the values"],  
For[i = 1, i <= Nmax, i++, m = (x0 + x1) / 2;  
If[Abs[(x1 - x0) / 2] < eps,  
Return[N[m]], Print[i, "th iteration value is :", N[m]];  
Print["estimated error in", i, "the iteration is:", N[(x1 - x0) / 2]];  
If[f[m] * f[x1] > 0, x1 = m, x0 = m]]];  
Print["Root is :", m] *  
Print["estimated error in", "the iteration is:", (x1 - x0) / 2]]
```

```
1th iteration value is :0.5
estimated error in1the iteration is:0.5
2th iteration value is :0.25
estimated error in2the iteration is:0.25
3th iteration value is :0.125
estimated error in3the iteration is:0.125
4th iteration value is :0.1875
estimated error in4the iteration is:0.0625
5th iteration value is :0.21875
estimated error in5the iteration is:0.03125
6th iteration value is :0.203125
estimated error in6the iteration is:0.015625
7th iteration value is :0.195313
estimated error in7the iteration is:0.0078125
8th iteration value is :0.199219
estimated error in8the iteration is:0.00390625
9th iteration value is :0.201172
estimated error in9the iteration is:0.00195313
10th iteration value is :0.202148
estimated error in10the iteration is:0.000976563
11th iteration value is :0.20166
estimated error in11the iteration is:0.000488281
12th iteration value is :0.201416
estimated error in12the iteration is:0.000244141
13th iteration value is :0.201538
estimated error in13the iteration is:0.00012207
```

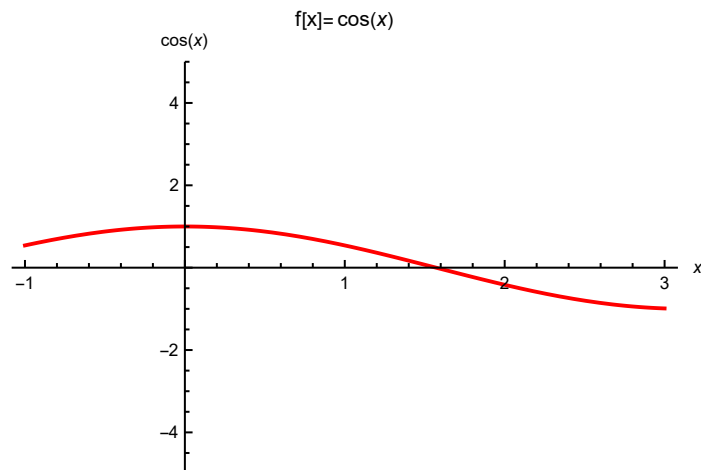
Out[*]=

```
Return[0.201599]
```

Question 2

```
f[x_] := Cos[x];
Plot[f[x], {x, -1, 3}, PlotRange -> {-5, 5},
PlotStyle -> {Red, Thick}, PlotLabel -> "f[x] = " f[x], AxesLabel -> {x, f[x]}]
```

Out[]:=



```
In[ ]:= x0 = 1;
x1 = 2;
Nmax = 20;
eps = 0.0001;
If[N[f[x0] * f[x1]] > 0,
Print["your value do not satisfy the IVP , so change the values"],
For[i = 1, i ≤ Nmax, i++, m = (x0 + x1) / 2;
If[Abs[(x1 - x0) / 2] < eps,
Return[N[m]], Print[i, "th iteration value is :", N[m]];
Print["estimated error in", i, "the iteration is:", N[(x1 - x0) / 2]];
If[f[m] * f[x1] > 0, x1 = m, x0 = m]]];
Print["Root is :", m] *
Print["estimated error in", "the iteration is:", (x1 - x0) / 2]
```

```
1th iteration value is :1.5
estimated error in1the iteration is:0.5
2th iteration value is :1.75
estimated error in2the iteration is:0.25
3th iteration value is :1.625
estimated error in3the iteration is:0.125
4th iteration value is :1.5625
estimated error in4the iteration is:0.0625
5th iteration value is :1.59375
estimated error in5the iteration is:0.03125
6th iteration value is :1.57813
estimated error in6the iteration is:0.015625
7th iteration value is :1.57031
estimated error in7the iteration is:0.0078125
8th iteration value is :1.57422
estimated error in8the iteration is:0.00390625
9th iteration value is :1.57227
estimated error in9the iteration is:0.00195313
10th iteration value is :1.57129
estimated error in10the iteration is:0.000976563
11th iteration value is :1.5708
estimated error in11the iteration is:0.000488281
12th iteration value is :1.57056
estimated error in12the iteration is:0.000244141
13th iteration value is :1.57068
estimated error in13the iteration is:0.00012207
```

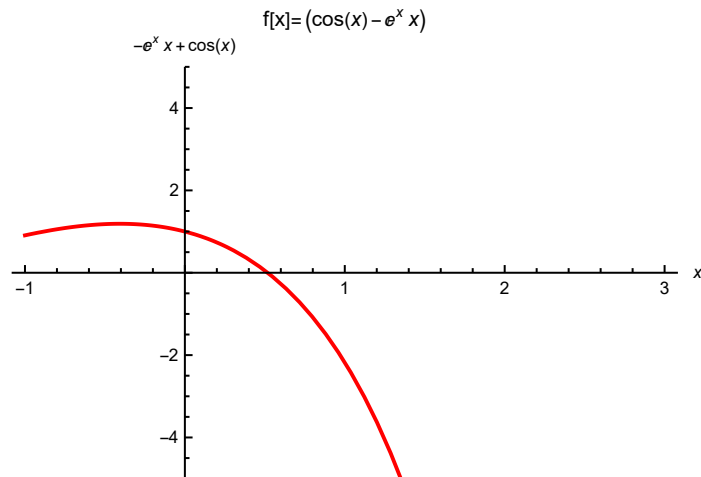
Out[*]=

```
Return[1.57074]
```

Question 3

```
In[*]:= f[x_] := Cos[x] - x * Exp[x];
Plot[f[x], {x, -1, 3}, PlotRange -> {-5, 5},
PlotStyle -> {Red, Thick}, PlotLabel -> "f[x] = " f[x], AxesLabel -> {x, f[x]}]
```

Out[*]=



```
In[*]:= x0 = 0;
x1 = 1;
Nmax = 20;
eps = 0.0001;
If[N[f[x0] * f[x1]] > 0,
Print["your value do not satisfy the IVP , so change the values"],
For[i = 1, i ≤ Nmax, i++, m = (x0 + x1) / 2;
If[Abs[(x1 - x0) / 2] < eps,
Return[N[m]], Print[i, "th iteration value is :", N[m]];
Print["estimated error in", i, "the iteration is:", N[(x1 - x0) / 2]];
If[f[m] * f[x1] > 0, x1 = m, x0 = m]]];
Print["Root is :", m] *
Print["estimated error in", "the iteration is:", (x1 - x0) / 2]]
```

```
1th iteration value is :0.5
estimated error in1the iteration is:0.5
2th iteration value is :0.75
estimated error in2the iteration is:0.25
3th iteration value is :0.625
estimated error in3the iteration is:0.125
4th iteration value is :0.5625
estimated error in4the iteration is:0.0625
5th iteration value is :0.53125
estimated error in5the iteration is:0.03125
6th iteration value is :0.515625
estimated error in6the iteration is:0.015625
7th iteration value is :0.523438
estimated error in7the iteration is:0.0078125
8th iteration value is :0.519531
estimated error in8the iteration is:0.00390625
9th iteration value is :0.517578
estimated error in9the iteration is:0.00195313
10th iteration value is :0.518555
estimated error in10the iteration is:0.000976563
11th iteration value is :0.518066
estimated error in11the iteration is:0.000488281
12th iteration value is :0.517822
estimated error in12the iteration is:0.000244141
13th iteration value is :0.5177
estimated error in13the iteration is:0.00012207
```

Out[*]=

```
Return[0.517761]
```